

Adaptive Attention Computation: Master Team Playbook

Consolidated 6-Member Action Plans, Individual Steps & Synchronization Gates

Target Horizon: 8.5 – 9.5 Hours	Team Structure: 6 Engineers & Researchers	Hardware Target: Dual Tesla T4 (16GB VRAM)
Baseline Model: Qwen/Qwen2.5-0.5B (FP16)	Core Deliverables: Kernels, CSVs, Deck & Report	Project Phase: Sprint Execution

1. How to Use This Execution Playbook

This master document contains the complete step-by-step action plans for all 6 members of Team 3. Work is organized using **Contract-First Engineering** so that every engineer can proceed in parallel without idle time. Each member has a dedicated 2-page action sheet detailing their mission, input/output contracts, hour-by-hour checklist, and operational guardrails. Either distribute the individual PDFs in team_action_plans/ or reference this consolidated guide.

2. Team Roster & Playbook Index

Role & Member	Core Discipline & Mission	Primary Output	Pages
Person 1: Theory Lead	Formulate error bound equations, input difficulty metric, and kernel math.	adaptive_kernel.py + Math Spec	Pages 2–3
Person 2: Compute Lead	Profile T4 latency, peak VRAM, and empirical scaling exponents (N=1K–64K).	01_T4_Efficiency.ipynb + CSVs	Pages 4–5
Person 3: Retrieval Lead	Implement Passkey retrieval suite; test Softmax/Linear/FAVOR+; run bootstrap CI.	02_Passkey_Suite.ipynb + Heatmaps	Pages 6–7
Person 4: Integration Lead	Monkey-patch Qwen2.5-0.5B attention blocks; benchmark prefill throughput.	patch_qwen.py + Prefill CSV	Pages 8–9
Person 5: Baselines Lead	Implement pure-PyTorch Gated DeltaNet layer; benchmark state retention.	03_DeltaNet_T4.ipynb + Table	Pages 10–11
Person 6: Synthesis Lead	Lock CSV schema; build automated chart generator; construct master 7-slide deck.	Master 7-Slide Deck + Executive Report	Pages 12–13

3. Team Synchronization Checkpoints & Handoff Pipeline

Checkpoint	Sprint Window	Synchronization Action & Mandatory Gate Requirements
SYNC 1: Interface Freeze	Hour 1.0	P6 locks unified CSV schema; P1 publishes mock kernel interface. All leads sign off.
SYNC 2: Baseline Check	Hour 3.5	P2 verifies Softmax scaling; P3 confirms Softmax 100% retrieval sanity check; P5 tests DeltaNet layer.
SYNC 3: Mid-Sprint Review	Hour 6.0	P2 presents linear scaling (9.7 ms at 64K); P3 presents retrieval collapse (14% at 1K). Pivot to adaptive kernel validated.
SYNC 4: Code Freeze	Hour 7.5	All code frozen. P1 hands real kernel to P4; P2, P3, P5 deliver CSVs to P6. No further code edits permitted.
SYNC 5: Deck Data Audit	Hour 8.5	P6 presents draft slide deck. Every lead audits their numbers against raw CSV files.
FINAL SIGN-OFF	Hour 9.5	15-minute presentation dry-run; lead Q&A; rehearsal; final deliverables submitted.

Sprint Action Plan: Person 1: Theory Lead

Mathematical Formulation & Adaptive Kernel Logic • Adaptive Attention Computation Sprint

Assigned Role: Person 1: Theory Lead	Estimated Workload: 9.0 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: Mathematical Modeling / Kernel Theory	Primary Handoff To: Person 4 (Model Patch) & Person 6 (Deck)	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Formulate the analytical error certificate linking positive-kernel approximation error to attention output accuracy. Design the dynamic rank selection rule $r^*(X, \delta)$ to balance computational efficiency against key-value retrieval quality. Verify numerical stability safeguards and provide mathematical rigor for all project claims.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Pre-registered hypothesis & variables	Team 3 Revised Proposal	Hour 0.0
INPUT	Empirical retrieval collapse numbers	Person 3 (Retrieval Lead)	Hour 6.0
OUTPUT	Mathematical error bound specification	Team Repository (theory_spec.md)	Hour 1.0
OUTPUT	Adaptive-rank kernel logic (adaptive_kernel.py)	Person 4 & Person 2	Hour 7.5
OUTPUT	Theory slide narratives (Slides 2, 5, 6)	Person 6 (Synthesis Lead)	Hour 8.0

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 1: Theory Lead:

- Error Certificate Condition: $\epsilon_r(X) \leq \delta * \min_i(Z_i) / [N * (2*V_{max} + \delta)]$
- Dynamic Rank Rule: $r^*(X, \delta) = \min\{ r \in \{64, 256, 1024\} : B(X, r) \leq \delta \}$
- Analytical Toy Check: $QK^T = [[1,2,3],[2,1,3],[1,1,2]]$; Linear output = [4.8, 5.2] vs Softmax = [4.28, 5.72]
- Kernel I/O Contract: `def adaptive_rank_attention(Q, K, V, delta=0.05, r_candidates=(64, 256, 1024), is_causal=False)`

4. Hour-by-Hour Master Action Checklist — Person 1: Theory Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Lock tensor I/O contract; publish mock kernel stub; finalize mathematical error bound formulation.	Frozen function signature + MockKernel	[] Done
1.0 – 3.0 h	Derive output error bounds; verify 3-token analytical toy example; compute exact relative Frobenius discrepancy.	Mathematical verification report	[] Done
3.0 – 5.5 h	Implement adaptive rank selection algorithm based on query-key entropy and logit temperature.	adaptive_kernel.py (Rank selection logic)	[] Done
5.5 – 7.5 h	Unit-test kernel for numerical stability, FP32 accumulator safeguards, and denominator epsilon clamping.	Unit test suite passing 100%	[] Done
7.5 – 8.5 h	Deliver production kernel to Person 4; review mathematical writeup with Person 6.	Kernel handoff + theory appendix	[] Done
8.5 – 9.5 h	Rehearse mathematical Q&A; defense for departmental presentation; verify final report equations.	Presentation sign-off	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
FP16 Accumulator Overflow	Enforce FP32 accumulators for all summations across N to prevent Inf/NaN in long context.
Zero Denominator	Clamp denominator with $\phi(Q) @ \phi(K)^T 1 + 1e-6$ to guarantee mathematical stability.
Associativity Limits	Never claim $(QK^T)V = Q(K^TV)$ holds for softmax; clearly explain the kernel substitution.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	Analytical 3-token toy calculation numerically verified ([4.8, 5.2]).
[]	Error bound certificate formula validated against empirical sweeps.
[]	adaptive_kernel.py adheres strictly to the frozen function signature.
[]	Slide 2, 5, and 6 narratives and LaTeX equations reviewed and signed off.

Sprint Action Plan: Person 2: Compute Lead

T4 Latency Profiling, Peak VRAM & Scaling Exponents • Adaptive Attention Computation Sprint

Assigned Role: Person 2: Compute Lead	Estimated Workload: 9.5 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: CUDA Performance Benchmarking / Systems Profiling	Primary Handoff To: Person 6 (Synthesis Lead)	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Profile execution time, peak memory, and empirical scaling exponents on Kaggle Tesla T4 across sequence lengths $N = 64$ to $65,536$. Establish the crossover point between Softmax, SDPA, and Linear Attention while enforcing strict OOM guardrails.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Frozen CSV schema contract	Person 6 (Synthesis Lead)	Hour 1.0
INPUT	Linear and mock attention kernels	Person 1 (Theory Lead)	Hour 1.0
OUTPUT	T4 Efficiency Notebook (01_T4_Efficiency.ipynb)	Team Repository	Hour 7.5
OUTPUT	Raw latency and memory logs (efficiency_raw.csv)	Person 6 (Synthesis Lead)	Hour 7.5
OUTPUT	High-res efficiency scaling charts (Fig 1)	Person 6 (Synthesis Lead)	Hour 8.0

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 2: Compute Lead:

- SDPA Backend Flags: `torch.backends.cuda.enable_mem_efficient_sdp(True); enable_flash_sdp(False)`
- Timing Harness: Use CUDA Events (`torch.cuda.Event(enable_timing=True)`) with explicit synchronization
- Memory Tracking: Measure $(\text{max_memory_allocated}() - \text{base_memory}) / (1024^2)$ in MB
- Sweep Regimes: $N \in \{64, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, 65536\}$; $B=1, H=4, d=64$

4. Hour-by-Hour Master Action Checklist — Person 2: Compute Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Setup Kaggle dual T4 harness; configure CUDA event timing harness; lock CSV output schema.	Benchmarking harness template	[] Done
1.0 – 3.0 h	Benchmark Naive Softmax vs SDPA across N = 64 to 32,768; record exact OOM boundary at N = 32,768.	Softmax & SDPA latency/memory curves	[] Done
3.0 – 5.5 h	Benchmark ReLU+1 Linear Attention latency and peak VRAM up to N = 65,536; confirm linear scaling slope.	Linear attention scaling curves	[] Done
5.5 – 7.5 h	Fit empirical scaling exponents (log-log slope α); calculate median speedups relative to Softmax and SDPA.	efficiency_raw.csv + scaling exponents	[] Done
7.5 – 8.5 h	Export publication-quality 3-panel scaling figures (latency, memory, speedup) to Person 6.	Rendered PNG charts delivered to P6	[] Done
8.5 – 9.5 h	Verify end-to-end Kaggle reproduction script executes without interactive timeouts.	Verified Kaggle reproduction notebook	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
CUDA OOM Crashes	Enforce strict batch size B=1 for all sequence lengths $N \geq 8,192$.
Asynchronous Timing Distortion	Always call torch.cuda.synchronize() before start and after end events.
Kaggle 60-min Idle Timeout	Submit long sweeps ($N \geq 16,384$) via 'Save & Run All (Commit)'.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	Latency measured up to N = 65,536 (confirming 9.74 ms for linear attention).
[]	Naive softmax OOM point documented at N = 32,768.
[]	efficiency_raw.csv populated with median, Q25, Q75, and peak VRAM.
[]	Scaling exponents α reported (0.80 for linear vs 1.81 for softmax).

Sprint Action Plan: Person 3: Retrieval Lead

Passkey Suite, Needle-in-a-Haystack & Statistical Non-Inferiority • Adaptive Attention Computation Sprint

Assigned Role: Person 3: Retrieval Lead	Estimated Workload: 9.0 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: Attention Quality Evaluation / Synthetic Probing	Primary Handoff To: Person 6 (Synthesis Lead)	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Evaluate long-range information retrieval and key-value binding fidelity using synthetic distractor suites. Quantify the retrieval accuracy collapse of linear attention and execute pre-registered bootstrap non-inferiority tests.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Frozen CSV schema contract	Person 6 (Synthesis Lead)	Hour 1.0
INPUT	Linear and FAVOR+ attention implementations	Person 1 (Theory Lead)	Hour 1.0
OUTPUT	Retrieval benchmark notebook (02_Passkey_Suite.ipynb)	Team Repository	Hour 7.5
OUTPUT	Retrieval accuracy CSVs (retrieval_raw.csv)	Person 6 (Synthesis Lead)	Hour 7.5
OUTPUT	Bootstrap non-inferiority audit table & heatmaps	Person 6 (Synthesis Lead)	Hour 8.0

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 3: Retrieval Lead:

- Class Coding: 16 balanced orthonormal classes via QR decomposition (DV=64, C=16); chance level = 6.25%
- Distractor Protocol: Query is target key with noise (noise=0.3); all other N-1 keys act as distractors
- Non-Inferiority Margin: $\delta = 0.05$; Paired difference $\Delta = \text{acc_linear} - \text{acc_softmax}$ across shared random seeds
- Resampling: 2,000 bootstrap iterations over independent retrieval tasks to construct 95% Confidence Intervals

4. Hour-by-Hour Master Action Checklist — Person 3: Retrieval Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Implement synthetic Passkey token generator with fixed-dimensional orthonormal class codes.	Passkey data generation harness	[] Done
1.0 – 3.0 h	Run baseline Softmax retrieval grid across N = 64 to 4,096; verify 100% accuracy sanity check.	Softmax baseline validation (100% acc)	[] Done
3.0 – 5.5 h	Execute Linear Attention (ReLU+1) Passkey sweep; document collapse from 57.7% (N=64) to 14.2% (N=1024).	Linear attention retrieval collapse curve	[] Done
5.5 – 7.5 h	Benchmark FAVOR+ sweeps across $r \in \{64, 256, 1024\}$ and key norm scales (2 to 24); test local vs global targets.	FAVOR+ rescue attempt data table	[] Done
7.5 – 8.5 h	Run 2,000 bootstrap iterations; compute 95% CI for Δ ; export noninferiority.csv and 2D heatmaps to Person 6.	noninferiority.csv + Fig 2 & Fig 3 charts	[] Done
8.5 – 9.5 h	Audit retrieval figures and non-inferiority intervals in presentation deck; participate in dry-run.	Presentation audit sign-off	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
One-Hot Dimension Inflation	Never use N-dimensional one-hot values; increasing d_v with N invalidates scaling comparisons.
Weak Baseline logit scale	Verify softmax achieves $\geq 95\%$ accuracy on baseline tasks before interpreting relative drops.
Permutation Equivariance	Control key distractor counts separately from target token sequence position.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	Baseline sanity check verified: Softmax achieves 100.0% accuracy on retrieval tasks.
[]	Linear collapse accurately documented (14.2% at N=1024, 9.5% at N=4096).
[]	Bootstrap 95% CI calculated, confirming rejection of non-inferiority ($\Delta = -0.86$ to -0.91).
[]	retrieval_raw.csv and noninferiority.csv delivered to Person 6.

Sprint Action Plan: Person 4: Integration Lead

Qwen2.5-0.5B Monkey-Patching, Prefill Throughput & Model Forward Pass • Adaptive Attention Computation Sprint

Assigned Role: Person 4: Integration Lead	Estimated Workload: 9.5 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: Deep Learning Frameworks / Pretrained Model Engineering	Primary Handoff To: Person 6 (Synthesis Lead)	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Integrate custom linear and adaptive attention kernels into a pretrained Transformer model (Qwen/Qwen2.5-0.5B). Validate end-to-end forward passes, benchmark prefill latency on Kaggle T4, and ensure memory footprints remain within 16GB.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Qwen/Qwen2.5-0.5B Hugging Face weights	Hugging Face Hub	Hour 0.0
INPUT	Mock kernel interface	Person 1 (Theory Lead)	Hour 1.0
INPUT	Production adaptive attention kernel	Person 1 (Theory Lead)	Hour 7.5
OUTPUT	Model monkey-patch script (patch_qwen.py)	Team Repository	Hour 7.5
OUTPUT	Prefill latency benchmark logs (qwen_prefill.csv)	Person 6 (Synthesis Lead)	Hour 8.0
OUTPUT	Patched model forward pass verification logs	Team Repository	Hour 8.5

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 4: Integration Lead:

- Model Baseline: Qwen/Qwen2.5-0.5B loaded in torch.float16 (~1.0 GB baseline VRAM footprint)
- Target Module: Monkey-patch Qwen2Attention.forward() in all transformer layers
- I/O Interface: Accept hidden_states (B, N, D), attention_mask, position_ids, past_key_values
- Sweep Sequences: Benchmark prefill latency across prompt lengths $N \in \{1024, 2048, 4096, 8192\}$

4. Hour-by-Hour Master Action Checklist — Person 4: Integration Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Setup Hugging Face environment on Kaggle T4; verify Qwen2.5-0.5B baseline FP16 inference.	Baseline inference verification	[] Done
1.0 – 3.0 h	Construct MockAttention monkey-patch harness to intercept and replace Qwen2Attention blocks.	patch_qwen.py (Monkey-patch infrastructure)	[] Done
3.0 – 5.5 h	Validate forward pass and tensor shapes using mock linear kernel; test generation loop stability.	Forward pass validation passing	[] Done
5.5 – 7.5 h	Profile prefill latency and peak VRAM across prompt lengths N = 1K to 8K comparing SDPA vs Linear.	qwen_prefill.csv benchmark logs	[] Done
7.5 – 8.5 h	Drop in Person 1's real AdaptiveRankAttention kernel; verify forward pass and prefill latency.	Patched model integration complete	[] Done
8.5 – 9.5 h	Push patched model codebase and verification scripts; participate in presentation rehearsal.	Repository codebase push & sign-off	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
Model Weight VRAM Bloat	Strictly load model in FP16 to preserve > 12 GB VRAM for long-context KV activations.
RoPE / Positional Incompatibility	Ensure rotary embeddings (RoPE) are applied to Q and K prior to linear kernel computation.
KV Cache Misalignment	Verify recurrent state accumulation matches autoregressive decoding requirements.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	patch_qwen.py monkey-patches Qwen2Attention cleanly without modifying library source.
[]	End-to-end forward pass succeeds on Qwen2.5-0.5B without NaNs or CUDA errors.
[]	Prefill latency benchmarked across N = 1K to 8K.
[]	VRAM usage stays strictly under 14 GB across all benchmark runs.

Sprint Action Plan: Person 5: Baselines Lead

SOTA Recurrent Baselines & Gated DeltaNet Implementation • Adaptive Attention Computation Sprint

Assigned Role: Person 5: Baselines Lead	Estimated Workload: 8.5 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: Recurrent Transformer Architecture / State-Space Models	Primary Handoff To: Person 6 (Synthesis Lead)	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Implement a pure-PyTorch layer for Gated DeltaNet to evaluate modern error-correcting recurrent mechanisms. Benchmark throughput, recurrent state retention, and memory scaling against standard linear attention on Kaggle T4.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Gated DeltaNet mathematical formulation	Literature / Project Specification	Hour 0.0
INPUT	Frozen CSV schema contract	Person 6 (Synthesis Lead)	Hour 1.0
OUTPUT	DeltaNet benchmark notebook (03_DeltaNet_T4.ipynb)	Team Repository	Hour 7.5
OUTPUT	Comparative baseline metrics table (deltanet_comparison.csv)	Person 6 (Synthesis Lead)	Hour 8.0
OUTPUT	Slide 7 SOTA baseline summary notes	Person 6 (Synthesis Lead)	Hour 8.5

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 5: Baselines Lead:

- Delta Update Rule: $S_t = S_{t-1} + \beta_t * (v_t - S_{t-1} k_t) * k_t^T$
- Gating Mechanism: Data-dependent decay gate $\alpha_t \in (0, 1)$ ensuring stable memory retention
- Implementation Rule: Pure PyTorch layer (avoid custom Triton/C++ extensions to ensure Kaggle compatibility)
- Benchmarking Regimes: Profile latency and recurrent memory across $N \in \{1024, 2048, 4096, 8192, 16384\}$

4. Hour-by-Hour Master Action Checklist — Person 5: Baselines Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Extract Gated DeltaNet recurrent mathematics and define tensor dimension contracts.	DeltaNet mathematical specification	[] Done
1.0 – 3.0 h	Implement standalone pure-PyTorch layer for Gated DeltaNet with data-dependent update gates.	Standalone DeltaNet PyTorch layer	[] Done
3.0 – 5.5 h	Profile DeltaNet throughput and execution latency on Kaggle T4 across N = 1K to 16K.	DeltaNet throughput profiling data	[] Done
5.5 – 7.5 h	Profile recurrent state memory footprint vs Linear Attention; evaluate state retention on associative recall.	State retention & memory comparison	[] Done
7.5 – 8.5 h	Finalize comparative baseline metrics table; export deltanet_comparison.csv to Person 6.	deltanet_comparison.csv delivered to P6	[] Done
8.5 – 9.5 h	Audit comparative SOTA claims in Slide 7; participate in presentation rehearsal.	Slide 7 audit sign-off	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
Compilation Failures	Strictly implement DeltaNet in pure PyTorch (torch.matmul / einsum) without custom CUDA extensions.
Recurrent Loop Latency	Use chunked associative scans where possible or batch sequence iterations to optimize T4 throughput.
Numerical Divergence	Bound delta updates with sigmoid / tanh activations on beta gates to avoid exploding state norms.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	Standalone Gated DeltaNet layer functions cleanly in pure PyTorch.
[]	Throughput benchmarked up to N = 16,384 on Tesla T4.
[]	deltanet_comparison.csv populated with latency, memory, and state retention metrics.
[]	Comparative analysis delivered to Person 6 for Slide 7.

Sprint Action Plan: Person 6: Synthesis Lead

Sprint Coordination, Automated Chart Generator & Master Presentation Deck • Adaptive Attention Computation Sprint

Assigned Role: Person 6: Synthesis Lead	Estimated Workload: 9.5 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: Technical Synthesis / Project Leadership & Presentation Design	Primary Handoff To: Departmental Reviewers / Stakeholders	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Enforce sprint-wide synchronization, lock data logging schemas, build the automated publication chart generator, and assemble the master 7-slide departmental presentation deck and final executive technical report.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Efficiency CSVs & exponents	Person 2 (Compute Lead)	Hour 7.5
INPUT	Retrieval CSVs & 95% CIs	Person 3 (Retrieval Lead)	Hour 7.5
INPUT	Prefill latency logs	Person 4 (Integration Lead)	Hour 8.0
INPUT	DeltaNet baseline tables	Person 5 (Baselines Lead)	Hour 8.0
INPUT	Theoretical derivations & narrative	Person 1 (Theory Lead)	Hour 8.0
OUTPUT	Unified CSV schema specification	All Team Members	Hour 1.0
OUTPUT	Automated chart generator script (plot_results.py)	Team Repository	Hour 3.0
OUTPUT	Master 7-Slide Deck (Team3_Attention_Results.pptx)	Project Stakeholders	Hour 9.0
OUTPUT	Consolidated Executive Technical Report (PDF)	Project Stakeholders	Hour 9.5

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 6: Synthesis Lead:
• CSV Contract: [timestamp, sequence_length_N, method, metric_name, metric_value, peak_vram_mb, status] • Slide Structure: 7 slides (1: Dilemma, 2: Hypothesis, 3: Design, 4: Findings, 5: Gap, 6: Architecture, 7: Roadmap) • Chart Styling: Matplotlib publication style (log2 x-axis, log10 y-axis, 95% CI error bars, high-contrast palette) • Acceptance Criteria: Enforce $R^* \geq 2.0\times$ speedup AND paired non-inferiority $\Delta > -0.05$

4. Hour-by-Hour Master Action Checklist — Person 6: Synthesis Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Lock unified CSV schema; setup presentation repository; chair Hour 1.0 Interface Freeze sync.	Unified CSV schema + Repo setup	[] Done
1.0 – 3.0 h	Build automated chart generation script with mock data templates; structure 7-slide deck master layout.	plot_results.py + 7-slide deck template	[] Done
3.0 – 5.5 h	Draft slide narratives and speaker notes for Slides 1, 2, 3, 4, 5; monitor mid-sprint progress.	Draft slide narratives & speaker notes	[] Done
5.5 – 7.5 h	Ingest real CSVs from P2, P3, P4, P5; auto-render final publication charts; enforce Code Freeze at 7.5h.	Rendered final publication charts	[] Done
7.5 – 8.5 h	Assemble master slide deck; audit every data point on slides against raw CSV logs; format executive report.	Master deck assembled + audited	[] Done
8.5 – 9.5 h	Chair presentation dry-run; manage rehearsal Q&A; obtain final deliverable sign-offs from all leads.	Final Presentation Deck + Signed Report	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
Data Inconsistency	Every numerical claim on slides must trace back to an exact row in the raw CSV files.
Slide Text Overcrowding	Enforce maximum 4 bullet points per slide; use visual charts for data density.
Unfavorable Control Omission	Must present PyTorch SDPA exact attention alongside Linear Attention on all scaling charts.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	Unified CSV schema adopted by all members at Hour 1.0.
[]	All empirical charts auto-rendered from real benchmark CSVs.
[]	Master 7-slide deck fully assembled with speaker notes and audited numbers.
[]	Consolidated executive report signed off and published.